COURSE 6 · BIOLOGY

Biology

Visual companion and Symbols Sheets

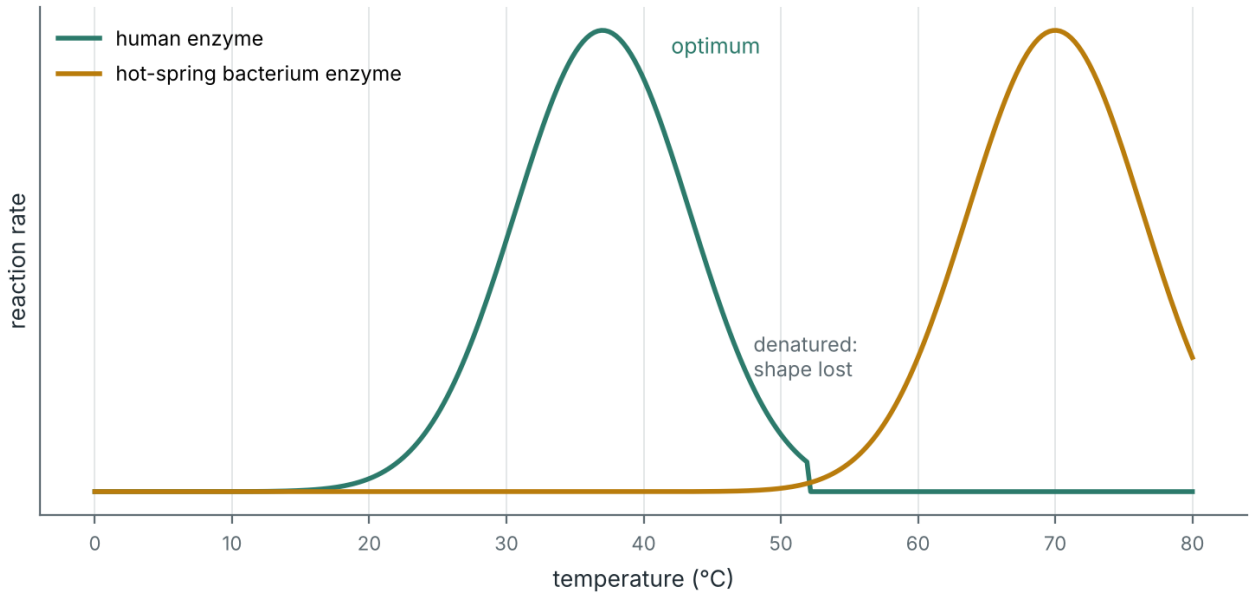
One section per module: the Symbols Sheet, with every numbered equation from the audio reader in ordinary notation, then the module's key diagram.

Keep it open while you listen, and beside you for the workbook.

Module 6.1 · Symbols and diagrams

The chemistry of life

6.1-1	Condensation (making a polymer)	$\text{monomer} + \text{monomer} \rightarrow \text{dimer} + \text{H}_2\text{O}$
6.1-2	Enzyme kinetics (Michaelis–Menten)	$v = \frac{V_{\max}[S]}{K_m + [S]}$
6.1-3	Enzyme reaction scheme	$E + S \rightleftharpoons ES \rightarrow E + P$

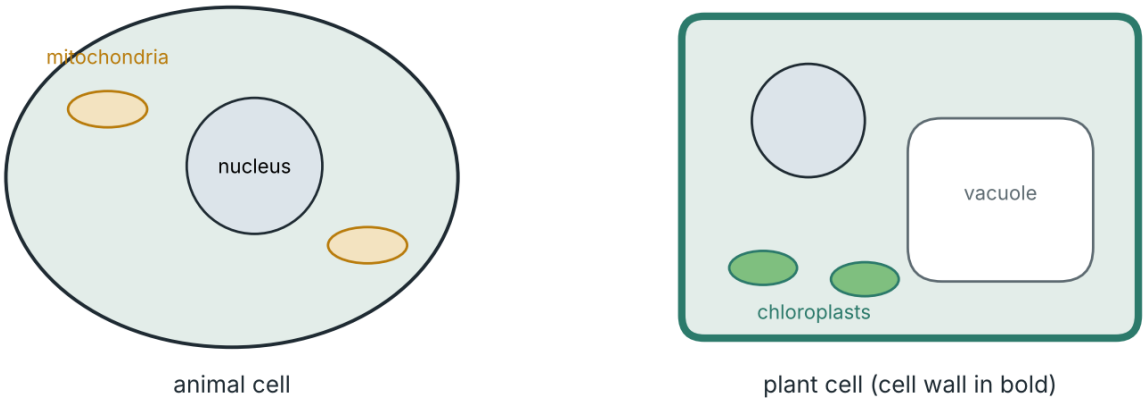


Enzyme activity rises with temperature to an optimum, then collapses as the protein denatures. Enzymes from hot-spring bacteria have much higher optima.

Module 6.2 · Symbols and diagrams

Cells

6.2-1	Surface-to-volume ratio of a cube of side L	$\frac{SA}{V} = \frac{6L^2}{L^3} = \frac{6}{L}$
6.2-2	Magnification	$M = \frac{\text{image size}}{\text{actual size}}$
6.2-3	Fick's law (diffusion rate)	$\text{rate} \propto \frac{A\,\Delta c}{d}$



animal cell

plant cell (cell wall in bold)

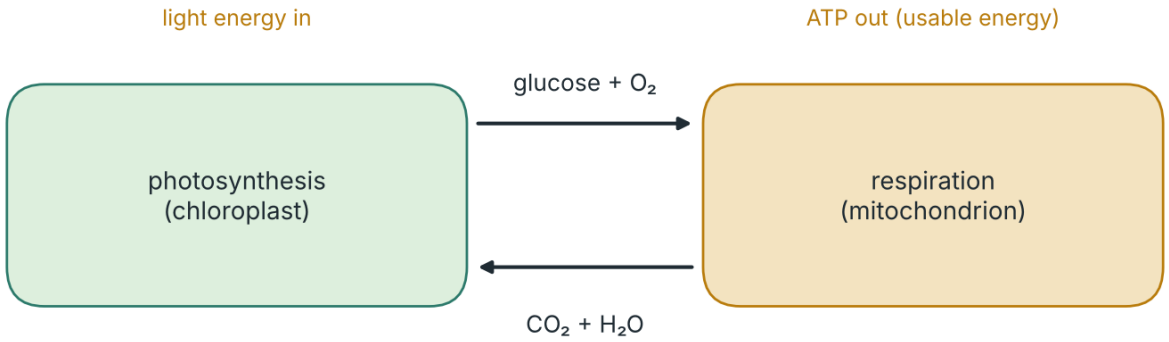
Simplified animal and plant cells.

Organelle	Job
nucleus	holds DNA, controls the cell
mitochondrion	aerobic respiration, makes ATP
ribosome	builds proteins
chloroplast (plants)	photosynthesis
cell membrane	controls what enters and leaves
cell wall (plants, bacteria)	support and protection

Module 6.3 · Symbols and diagrams

Metabolism

6.3-1	ATP hydrolysis	$\mathrm{ATP + H_2O \rightarrow ADP + P_i + energy}$
6.3-2	Photosynthesis	$\mathrm{6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2}$
6.3-3	Aerobic respiration	$\mathrm{C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O\ (\approx 30\ ATP)}$
6.3-4	Fermentation in yeast	$\mathrm{C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2}$



Photosynthesis and respiration are mirror images, cycling carbon, oxygen and water.

Module 6.4 · Symbols and diagrams

DNA, genes and protein synthesis

6.4-1	Chargaff's rule (from base pairing)	$[A] = [T], \qquad [G] = [C]$
6.4-2	Central dogma	$\mathrm{DNA} \rightarrow \mathrm{RNA} \rightarrow \mathrm{protein}$
6.4-3	Number of codons	$4^3 = 64$





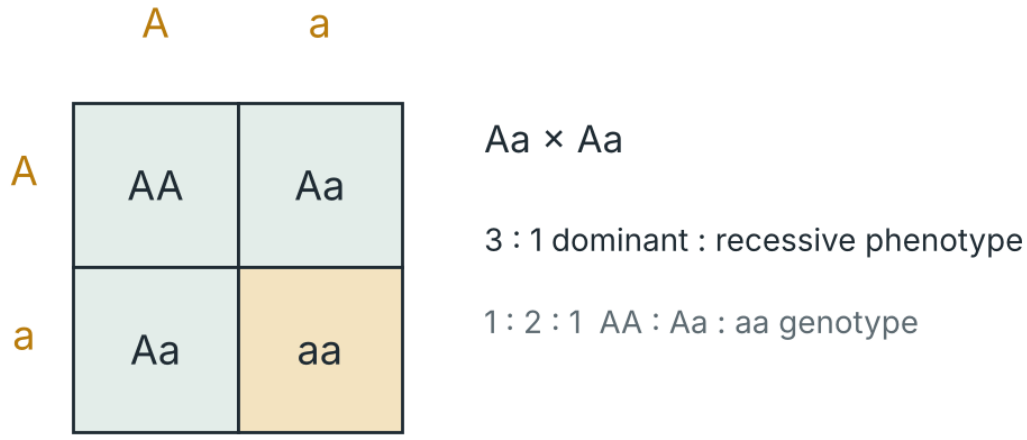
Each three-letter codon in mRNA specifies one amino acid.

Complementary base pairing, and the flow of information from DNA to protein.

Module 6.5 · Symbols and diagrams

Inheritance and genetics

6.5-1	Monohybrid cross ratios	$Aa \times Aa: \quad 1\,AA : 2\,Aa : 1\,aa$
6.5-2	Chromosome combinations from independent assortment	$2^{23} \approx 8.4 \times 10^{6}$
6.5-3	Hardy–Weinberg (allele frequencies p, q)	$p^2 + 2pq + q^2 = 1$

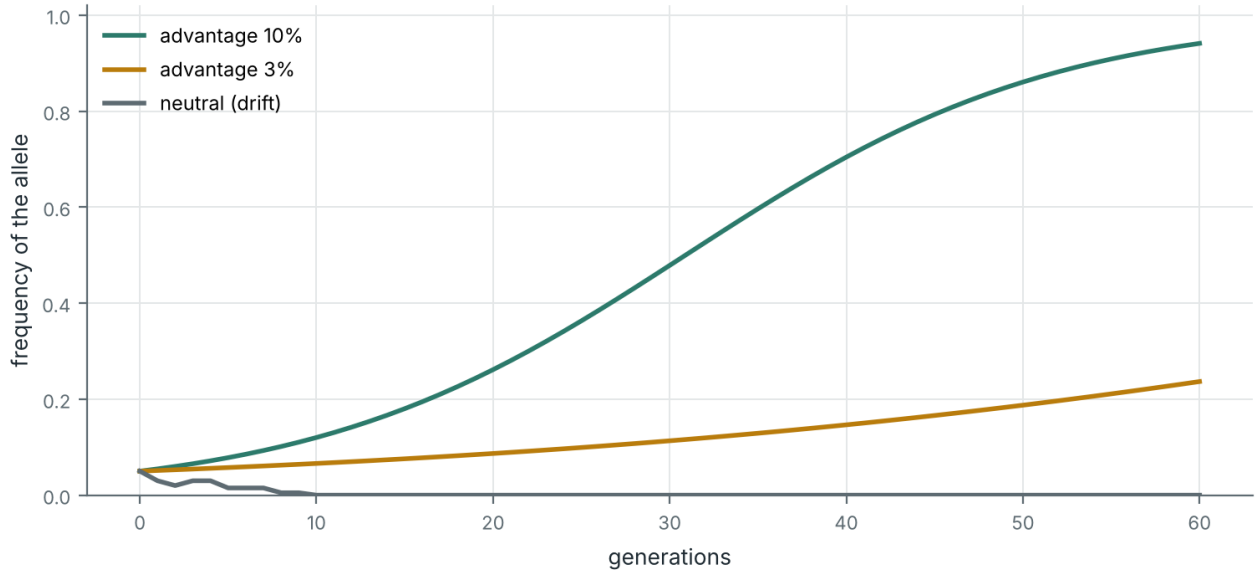


Punnett square for a cross between two heterozygotes.

Module 6.6 · Symbols and diagrams

Evolution and natural selection

6.6-1	Allele frequency change under selection	$p' = \frac{p(1+s)}{p(1+s)+(1-p)}$
6.6-2	Relative fitness	$w = \frac{\text{offspring of a type}}{\text{offspring of the best type}}$

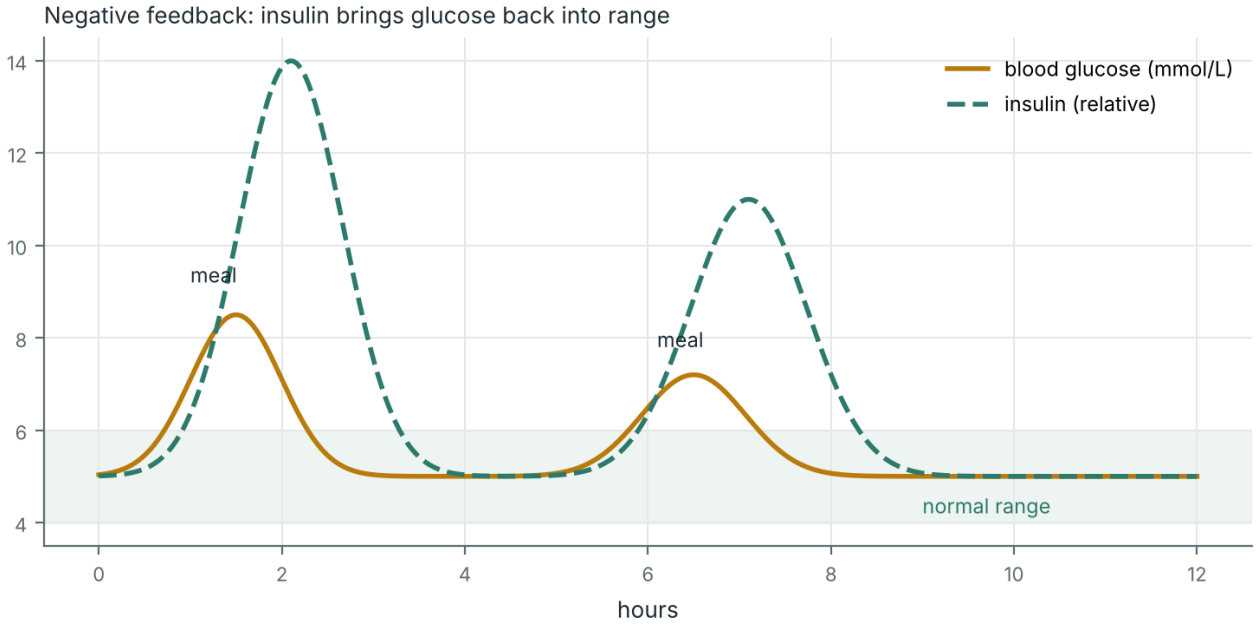


Simulated spread of an allele starting at 5%: a 10% advantage takes over within decades of generations; a neutral allele wanders by chance.

Module 6.7 · Symbols and diagrams

Human body systems

6.7-1	Cardiac output	$\mathrm{CO} = \mathrm{HR} \times \mathrm{SV}$
6.7-2	Body mass index (a population screening tool)	$\mathrm{BMI} = \frac{m\,(\mathrm{kg})}{h^{2}\,(\mathrm{m}^{2})}$

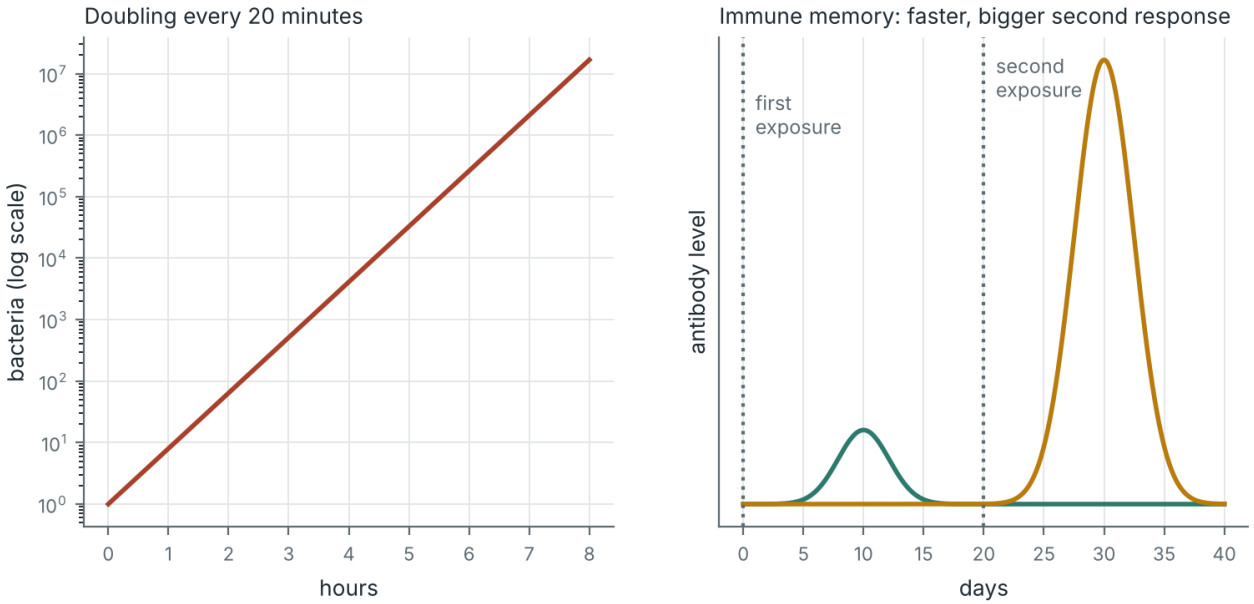


Blood glucose and insulin after two meals: negative feedback returns glucose to its normal range.

Module 6.8 · Symbols and diagrams

Microbes, immunity and disease

6.8-1	Bacterial growth	$N = N_0 \times 2^{t/t_d}$
6.8-2	Herd immunity threshold	$H = 1 - \frac{1}{R_0}$
6.8-3	Effective reproduction number	$R_{\text{eff}} = R_0 \times (\text{fraction susceptible})$

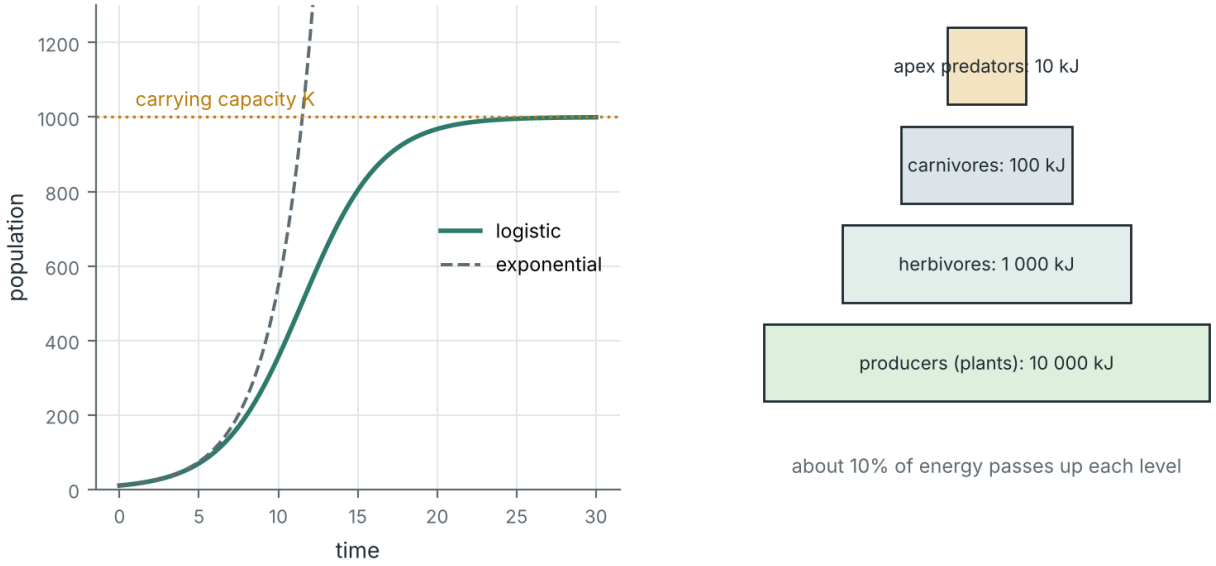


Left: unchecked bacterial growth. Right: immune memory gives a faster, larger response on second exposure.

Module 6.9 · Symbols and diagrams

Ecology and ecosystems

6.9-1	Ecological efficiency	$E_{n+1} \approx 0.1\, E_n$
6.9-2	Logistic growth	$\frac{dN}{dt} = rN\left(1 - \frac{N}{K}\right)$
6.9-3	Exponential growth	$N = N_0 e^{rt}$



Left: logistic versus exponential growth. Right: an energy pyramid, about 10% transferred per level.